

Exploratory Analysis of a 50,000-Patient Synthetic Diabetes Risk Dataset: Strong Univariate Separability with Near-Absent Multivariate Structure

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Abstract—This work presents an exploratory analysis of a publicly available synthetic dataset of diabetes risk, with the goal of characterising which of its predictors carry discriminative information before any predictive model is considered. The dataset comprises $N = 50,000$ records, $D = 22$ predictors and $L = 3$ risk classes, with a strongly imbalanced class distribution in which the *Low* class holds only 0.94% of the observations. We combine unconditional and class-conditional univariate statistics, pairwise Pearson correlation and a principal component analysis (PCA) implemented from first principles, without recourse to pre-made PCA routines. Two glycemic markers, fasting blood sugar and HbA1c, separate the risk classes almost by themselves, yet the first two principal components account for only 15.2% of the total variance. Three independent lines of evidence, an excess kurtosis of -1.2 in 19 of the 20 quantitative predictors, a median absolute pairwise correlation of 0.003, and an eigenvalue spectrum flat at unity, converge on the same conclusion: the predictors were drawn independently from bounded uniform distributions, and the only structural relationship the data contain is the algebraic identity that defines the body mass index.

Index Terms—exploratory data analysis, principal component analysis, synthetic data, diabetes risk, class separability

I. INTRODUCTION

Diabetes mellitus is one of the largest public health burdens worldwide [1]. In 2024, 589 million adults aged 20 to 79 years, one in every nine, were living with the condition, and this figure is projected to reach 853 million by 2050, a 45% rise against a projected population growth of 25%. A further 252 million adults are estimated to be unaware that they have it. The disease accounted for 3.4 million deaths and for at least USD 1 trillion in health expenditure in that single year, which is what makes early detection consequential: it opens a window for treatment and lifestyle intervention before microvascular and cardiovascular complications become established.

Fitting a classifier without a prior exploratory analysis makes it impossible to tell whether the performance obtained reflects genuine clinical patterns or artefacts of the data. Descriptive statistics and linear projections expose, beforehand,

redundancy among predictors, disparities of scale, outliers and the degree of overlap between the classes. Without that map of the underlying structure, neither the success nor the failure of a model can be diagnosed.

The use of synthetic datasets in health care has grown as a way of circumventing strict patient privacy constraints, of compensating for the scarcity of records and of enabling educational applications [2]. That convenience, however, carries the risk that the generating process fails to reproduce the complex physiological interdependencies present in real clinical populations. When multivariate relationships are not preserved, algorithms trained on such data tend to learn mathematical artefacts rather than legitimate biological mechanisms. It is in this context that the dataset analysed here belongs, its origin being explicitly synthetic [3].

This work presents an exploratory and multivariate characterisation of a synthetic dataset of diabetes risk, assessing its statistical consistency and its clinical fidelity before any modelling step. We investigate the dependencies among predictors and determine to what extent the risk classes are structurally separable, both in the original predictor space and under projections of maximum variance. The remainder of this paper is organised as follows. Section II describes the dataset and the analytical methods. Section III reports and discusses the results. Section IV concludes.

II. METHODS

A. Dataset

The dataset was obtained from Kaggle [3] and is explicitly described by its authors as a large-scale *synthetic* healthcare dataset. The raw file holds 41 columns, from which we discard `Patient_ID`, a row identifier carrying no physiological information, and `Diabetes_Risk_Score`, the continuous score from which the categorical label is derived. This leaves $N = 50,000$ observations and $D = 22$ predictors, of which 20 are quantitative and two (`Country` and `Gender`) are

categorical. The class label `Diabetes_Risk` takes $L = 3$ values.

Table I reports the class distribution. The disparity between the 36,593 observations of class *High* and the 470 of class *Low*, a ratio of 78 to 1 and a mere 0.94% of the sample, imposes structural limits on the analysis. It renders the class-conditional estimates of the minority class unstable, and it would compromise the evaluation of any future classifier, since a naive model that constantly predicts *High* would already reach 73% accuracy.

TABLE I
CLASS DISTRIBUTION OF `DIABETES_RISK`

Class	N_l	Share
High	36,593	73.19%
Moderate	12,937	25.87%
Low	470	0.94%
Total	50,000	100.00%

Table II summarises the missing values, which affect 12 of the 20 quantitative predictors. Four of them are missing *exactly* one thousand values each, and those four are precisely the complete lipid panel: total cholesterol, HDL, LDL and triglycerides. A fifth column outside the quantitative set, medication adherence, shows the same round count. Natural record loss does not produce an identical round figure across five variables, so the absences were evidently injected by the generator. They were not injected by examination, however: no observation lacks all four lipid measurements at once, and the pairwise overlap between them ranges from 12 to 21 rows, which is what independent sampling of 2% of the records would yield. In clinical practice a lipid panel is a single blood draw and its four results go missing together; here a fixed quota of one thousand absences was drawn column by column, ignoring that dependency.

TABLE II
MISSING VALUES (ONLY AFFECTED PREDICTORS SHOWN)

Predictor	Missing
Weight kg	2,962
Height cm	2,945
Sleep Hours	2,017
Exercise Hours Per Week	1,992
HbA1c	1,974
Daily Walking Minutes	1,964
LDL	1,000
HDL	1,000
Total Cholesterol	1,000
Triglycerides	1,000
Blood Glucose	959
Age	497

B. Pre-processing

Standardising the variables to zero mean and unit variance before the principal component analysis is mandatory, since the raw predictors operate on widely disparate scales (mg/dL,

kg, cm, years); without that correction the variables of larger magnitude would dominate the covariance matrix and the analysis would describe the units of measurement rather than the intrinsic structure of the data. Furthermore, because the covariance matrix requires complete records, the rows holding missing values were removed. This exclusion reduced the set from 50,000 to 33,689 observations, that is, 32.6% of the data discarded, and it establishes a methodological limitation: the multivariate analysis describes a complete subset and not the entire cohort. The univariate and bivariate statistics, in contrast, use every observation available for each predictor.

C. Analytical methods

For each predictor d we compute the mean μ_d , the standard deviation σ_d and the skewness γ_d ,

$$\gamma_d = \frac{1}{N} \sum_{n=1}^N \left(\frac{x_{n,d} - \mu_d}{\sigma_d} \right)^3, \quad (1)$$

which quantifies the imbalance between the tails of a distribution and vanishes for any density symmetric about its mean, together with the excess kurtosis, the analogous fourth-order moment minus three, and their class-conditional counterparts $\mu_{d|l}$, $\sigma_{d|l}$ and $\gamma_{d|l}$ over the N_l observations of each class l . Skewness vanishes for any symmetric distribution and so cannot by itself distinguish a Gaussian from a uniform density, whereas the excess kurtosis separates them cleanly: it is 0 for the Gaussian and $-6/5$ for the uniform. We therefore report both.

Linear dependence between pairs of predictors is quantified by the Pearson correlation coefficient

$$\rho_{d_i, d_j} = \frac{\mathbb{E} [(X_{d_i} - \mu_{d_i})(X_{d_j} - \mu_{d_j})]}{\sigma_{d_i} \sigma_{d_j}}, \quad (2)$$

which captures linear association only. A vanishing ρ does not imply independence in general, so we treat the correlation matrix as one piece of evidence among three rather than as a conclusion.

To rank predictors by their ability to separate classes on their own, we use the ratio between the range of the class-conditional means and the unconditional standard deviation,

$$\delta_d = \frac{\max_l \mu_{d|l} - \min_l \mu_{d|l}}{\sigma_d}, \quad (3)$$

so that δ_d expresses the class separation in units of the predictor's own spread and is comparable across predictors measured in different physical units.

Finally, the principal component analysis was implemented from scratch, without pre-made PCA functions or libraries, following the standard chain [4]: standardisation, estimation of the covariance matrix $\mathbf{S}$, eigendecomposition $\mathbf{S} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$, and projection of the observations onto the two eigenvectors of largest eigenvalue. The only library routine used is the symmetric eigensolver.

III. RESULTS AND DISCUSSION

A. Unconditional univariate analysis

Table III reports μ_d , σ_d , γ_d , the excess kurtosis and the observed range of the 20 quantitative predictors, and Fig. 1 shows the corresponding histograms.

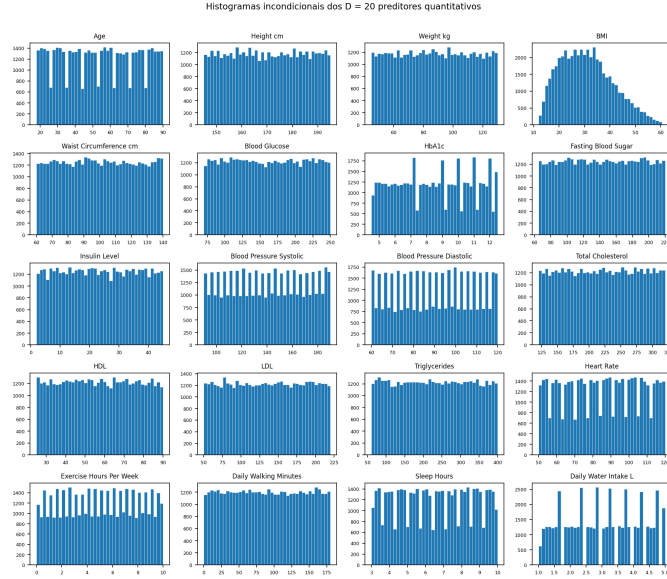


Fig. 1. Unconditional histograms of the 20 quantitative predictors. All but the body mass index are visibly flat-topped.

TODO: comentar que todas as assimetrias sao praticamente nulas, entre -0.015 e $+0.008$, com a unica excecao do IMC ($\gamma = 0.432$). Simetria, porem, nao implica normalidade, e e a curtose que resolve a questao: 19 dos 20 preditores tem curtose excedente entre -1.21 e -1.19 , indistinguível do valor $-6/5$ de uma distribuicao uniforme, e os histogramas da Fig. 1 sao visivelmente achatados e truncados. Os limites reforcam a leitura, pois sao numeros redondos escolhidos por conveniencia e nao valores fisiologicos: idade de 18 a 90 anos, frequencia cardiaca de 50 a 120 bpm, triglicerideos de 60 a 400 mg/dL, sono de 3 a 10 horas.

TODO: a unica excecao e o IMC, com curtose -0.45 e assimetria positiva, e a excecao tem explicacao mecanica: sendo o quociente peso/altura² de duas uniformes independentes, ele nao e uniforme. O IMC nao foi sorteado, foi *calculado*. Este e o primeiro dos tres sinais que convergem na Secao III-D.

B. Class-conditional analysis and discriminative power

Table IV reports the class-conditional means and ranks the predictors by the discriminative power of (3); Fig. 2 shows the class-conditional histograms and Fig. 3 the box-plots of the four strongest predictors.

TODO: responder diretamente a pergunta do enunciado. Sim, ha preditores com poder discriminativo isolado, e sao dois: glicose em jejum ($\delta = 1.44$) e HbA1c ($\delta = 1.38$), seguidos a distancia por idade (0.84), IMC (0.61) e peso

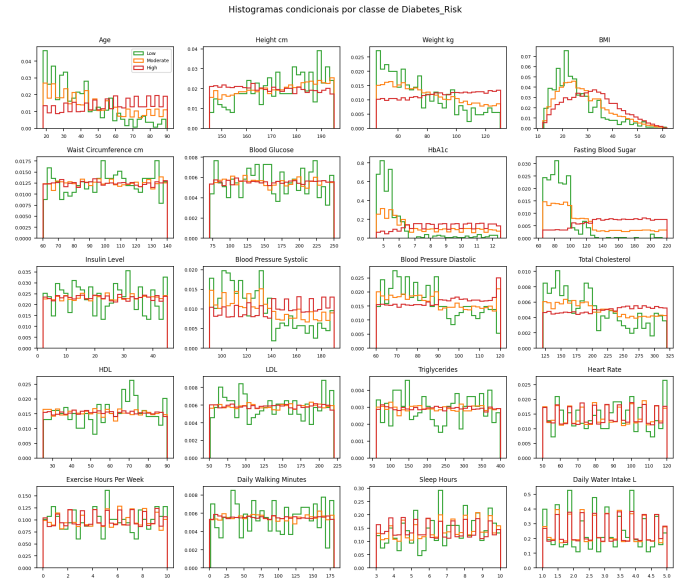


Fig. 2. Class-conditional histograms of the 20 quantitative predictors, normalised to unit area.

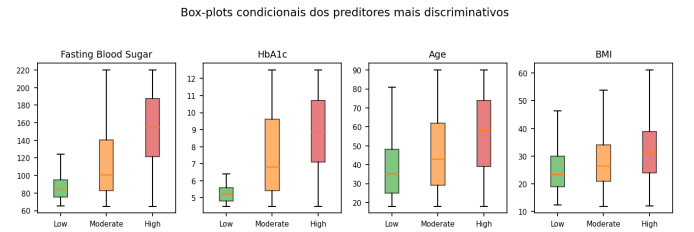


Fig. 3. Class-conditional box-plots of the four most discriminative predictors.

(0.57). Abaixo de $\delta \approx 0.1$ os onze preditores restantes nao separam nada.

TODO: HbA1c varia de 5.67 (Low) a 8.86 (High), exatamente a travessia da faixa clinica de normal para diabetico [5]; glicose em jejum vai de 88.4 a 152.8 mg/dL, atravessando o limiar diagnostico de 126; idade de 38.8 a 56.5 anos; IMC de 25.6 a 31.9, isto e, de sobrepeso para obesidade. Nesse sentido, os preditores que separam as classes sao clinicamente os corretos.

Three observations qualify this apparent clinical fidelity. TODO (i): a glicose casual tem $\delta = 0.020$, praticamente ruído, enquanto a glicose em jejum tem $\delta = 1.44$. Clinicamente as duas medem o mesmo substrato e caminham juntas, de modo que o gerador usou uma e ignorou a outra. TODO (ii): a circunferencia abdominal, marcador consagrado de risco metabolico, tem $\delta = 0.041$ e ainda por cima *decrece* com o risco, de 100.85 para 99.91 cm. TODO (iii): a altura decresce de 173.7 para 169.4 cm com $\delta = 0.30$, o que nao e fisiologia mas consequencia algebrica de o IMC dirigir o rotulo: a peso igual, quem e mais baixo tem IMC maior, e portanto a altura herda um sinal invertido que ela nao possui na populacao real.

TABLE III

UNCONDITIONAL UNIVARIATE STATISTICS OF THE 20 QUANTITATIVE PREDICTORS. THE EXCESS KURTOSIS OF A UNIFORM DISTRIBUTION IS $-6/5$.

Predictor	μ_d	σ_d	γ_d	Excess kurtosis	Range
Age	53.87	21.13	+0.008	-1.206	18.0–90.0
Height cm	170.02	14.46	+0.000	-1.207	145.0–195.0
Weight kg	87.46	24.50	+0.004	-1.191	45.0–130.0
BMI	30.92	10.29	+0.432	-0.453	11.9–61.7
Waist Circumference cm	99.96	23.09	+0.010	-1.191	60.0–140.0
Blood Glucose	159.91	52.04	+0.006	-1.212	70.0–250.0
HbA1c	8.49	2.31	+0.002	-1.204	4.5–12.5
Fasting Blood Sugar	142.61	44.68	-0.001	-1.200	65.0–220.0
Insulin Level	23.48	12.40	+0.006	-1.200	2.0–45.0
Blood Pressure Systolic	140.17	29.17	-0.004	-1.201	90.0–190.0
Blood Pressure Diastolic	90.04	17.58	-0.010	-1.193	60.0–120.0
Total Cholesterol	220.17	57.79	-0.005	-1.203	120.0–320.0
HDL	57.37	18.72	+0.004	-1.195	25.0–90.0
LDL	135.01	49.14	-0.004	-1.205	50.0–220.0
Triglycerides	229.54	98.23	-0.000	-1.200	60.0–400.0
Heart Rate	85.12	20.39	-0.015	-1.189	50.0–120.0
Exercise Hours Per Week	5.02	2.88	-0.009	-1.192	0.0–10.0
Daily Walking Minutes	89.93	52.03	+0.005	-1.206	0.0–180.0
Sleep Hours	6.50	2.03	-0.005	-1.210	3.0–10.0
Daily Water Intake L	3.00	1.15	+0.003	-1.189	1.0–5.0

TABLE IV
CLASS-CONDITIONAL MEANS AND DISCRIMINATIVE POWER δ_d ,
PREDICTORS SORTED BY δ_d

Predictor	$\mu_d _{\text{Low}}$	$\mu_d _{\text{Mod}}$	$\mu_d _{\text{High}}$	δ_d
Fasting Blood Sugar	88.44	115.81	152.78	1.44
HbA1c	5.67	7.54	8.86	1.38
Age	38.81	46.89	56.53	0.84
BMI	25.60	28.31	31.92	0.61
Weight kg	75.68	81.77	89.61	0.57
Blood Pressure Systolic	127.98	134.73	142.25	0.49
Total Cholesterol	200.41	211.82	223.37	0.40
Blood Pressure Diastolic	85.55	88.06	90.80	0.30
Height cm	173.66	171.76	169.35	0.30
Sleep Hours	6.78	6.66	6.44	0.17
HDL	58.68	57.25	57.40	0.076
Daily Water Intake L	2.94	2.98	3.01	0.059
Insulin Level	23.05	23.61	23.44	0.045
Waist Circumference cm	100.85	100.08	99.91	0.041
Daily Walking Minutes	91.20	90.24	89.81	0.027
Blood Glucose	159.01	159.46	160.07	0.020
Heart Rate	85.36	85.10	85.13	0.013
LDL	135.04	135.35	134.90	0.009
Exercise Hours Per Week	5.03	5.02	5.02	0.006
Triglycerides	229.47	229.47	229.57	0.001

C. Bivariate analysis

Fig. 4 shows the correlation matrix and Fig. 5 the pairwise scatter plots, coloured by class.

TODO: a matriz e essencialmente uma identidade. A mediana do modulo das correlacoes fora da diagonal e 0.003, e apenas dois pares fogem disso: IMC com peso ($\rho = +0.843$) e IMC com altura ($\rho = -0.516$), que sao a propria definicao do indice e nao um achado empirico. As nuvens da Fig. 5 sao retangulos sem estrutura, coerentes com preditores uniformes e independentes, e nenhum par separa visualmente as classes.

TODO: o achado central e que HbA1c e glicose em jejum, os dois unicos preditores com poder discriminativo relevante,

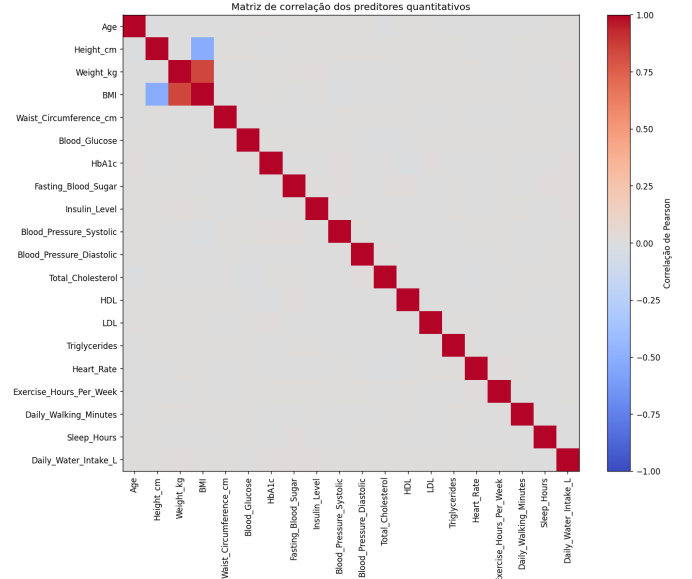


Fig. 4. Pairwise Pearson correlation matrix of the 20 quantitative predictors.

tem $\rho = -0.005$ entre si. Num paciente real esses dois exames sao fortemente correlacionados, pois a HbA1c integra a glicemia das semanas anteriores. Aqui o rotulo depende dos dois por caminhos independentes, o que e fisiologicamente impossivel e so se explica pelo processo gerador. Este e o segundo sinal.

D. Principal component analysis

Table V lists the eigenvalue spectrum and Fig. 6 shows the projection onto the first two components.

TODO: apos a padronizacao a soma dos autovalores e 20, de modo que um conjunto de preditores mutuamente nao correlacionados produz vinte autovalores iguais a 1. E quase

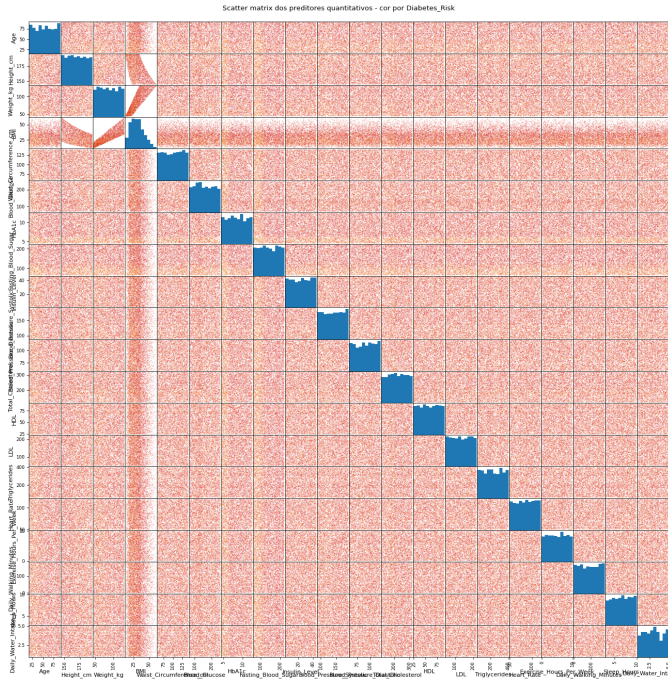


Fig. 5. Pairwise scatter plots of the quantitative predictors on a random subsample of 3,000 observations, coloured by risk class.

TABLE V
EIGENVALUE SPECTRUM OF THE STANDARDISED COVARIANCE MATRIX
(20 COMPONENTS, 33,689 COMPLETE OBSERVATIONS)

Component	λ	Explained	Cumulative
PC1	1.9943	9.97%	9.97%
PC2	1.0371	5.19%	15.16%
PC3	1.0327	5.16%	20.32%
PC4-PC19	0.9564–1.0292	—	—
PC20	0.0123	0.06%	100.00%

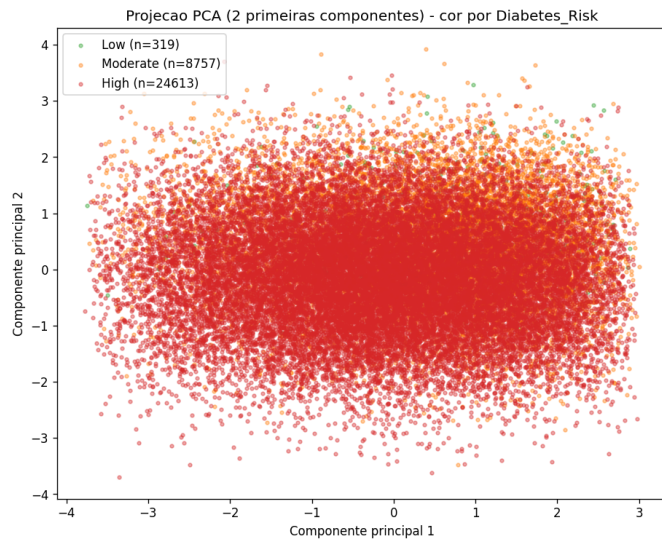


Fig. 6. Observations projected onto the first two principal components, coloured by risk class.

exatamente o que se observa: as componentes PC2 a PC19 ficam entre 0.956 e 1.037, uma faixa de menos de $\pm 4\%$ em torno da unidade. Um espectro plano e a assinatura da ausencia de estrutura linear, e constitui o terceiro sinal independente, vindo de um terceiro metodo, do que a curtose e a matriz de correlacao ja indicavam.

TODO: as duas excecoes sao a mesma coisa vista por dois lados. Os *loadings* da primeira componente ($\lambda = 1.994$) sao IMC -0.706 , peso -0.602 e altura $+0.372$, com todos os demais abaixo de 0.02 em modulo; os da ultima ($\lambda = 0.0123$) sao IMC $+0.708$, peso -0.603 e altura $+0.368$. O PCA isolou sozinho a identidade $IMC = \text{peso}/\text{altura}^2$: a maior variancia esta no plano gerado por essas tres variaveis, e o autovalor quase nulo mede a dependencia quase deterministica entre elas, quase e nao exatamente porque a relacao e nao linear e o PCA so captura sua melhor aproximacao linear. Esta e a unica relacao estrutural presente no conjunto, e o metodo a encontrou sem ser instruido.

TODO: responder as tres perguntas do item 5. (i) As classes nao ficam bem separadas na projecao, e ficam *pior* separadas do que na analise monovariada de HbA1c ou glicose em jejum. (ii) Nao ha fronteira linear visivel. (iii) *Moderate* se sobrepoe fortemente a *High*, e *Low* e praticamente invisivel, tanto por conter apenas 470 pontos quanto por ocupar a mesma regioao que as outras duas.

TODO: fechar com a tensao aparente. As duas primeiras componentes explicam apenas 15.2% da variancia e a projecao nao separa nada, enquanto a analise monovariada encontrou sinal forte e clinicamente coerente. As duas coisas sao verdadeiras ao mesmo tempo porque medem objetos diferentes: o PCA maximiza variancia, nao separabilidade, e quando os preditores sao quase todos independentes nao existe direcao de alta variancia onde a informacao de classe se concentre. Alem disso, a unica direcao de variancia elevada que o conjunto possui, a do IMC, e justamente uma relacao algebrica sem conteudo discriminativo proprio. O PCA nao falhou; ele reportou corretamente que nao ha estrutura multivariada a reportar.

IV. CONCLUSION

TODO: os tres achados convergem para uma leitura unica. A curtose uniforme em 19 de 20 preditores, a mediana de 0.003 nas correlacoes e o espectro de autovalores plano sao tres medidas independentes do mesmo fato: os preditores foram amostrados de distribuicoes uniformes independentes em intervalos redondos, e o rotulo foi atribuido a posteriori em funcao de poucas variaveis, principalmente glicose em jejum, HbA1c, idade e IMC. A unica dependencia presente e algebrica, nao fisiologica.

TODO: a implicacao pratica. Um dataset assim serve para exercitar o fluxo de analise, que e o objetivo deste trabalho, mas nao suporta conclusoes clinicas nem permite avaliar de forma realista metodos multivariados, justamente porque a estrutura multivariada que esses metodos existem para explorar foi eliminada na geracao. Um classificador treinado aqui alcançaria bom desempenho aparente usando dois preditores, e

esse resultado nao transferiria para dados reais. Registrar isso e, em si, um resultado da analise exploratoria.

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